

Dhan Vavhar FinOps Consulting

Google Cloud FinOps Cost Estimate Proposal

Project	Retail Inventory API
Prepared For	Retail Client Inc.
Prepared By	FinOps Advisory Team
Region	us-central1
Normalization Strategy	Performance
Date	2026-08-05
Estimate ID	29a7499f-7115-40c9-89d9-89d9dc390137

Executive Summary

This proposal presents a Google Cloud infrastructure estimate for **Retail Inventory API**, deployed in **us-central1**. The recommended configuration resolves to a total estimated cost of **USD 463.64 per month** (USD 5,563.68 per year). The request was validated against 11 rule(s), producing 3 warning(s) and 0 blocker(s). 2 configuration assumption(s) were made to map the request onto valid Google Cloud resources; each is documented in the Assumptions section below with full reasoning.

MONTHLY	YEARLY	3-YEAR
USD 463.64	USD 5,563.68	USD 16,691.04

Recommended Architecture

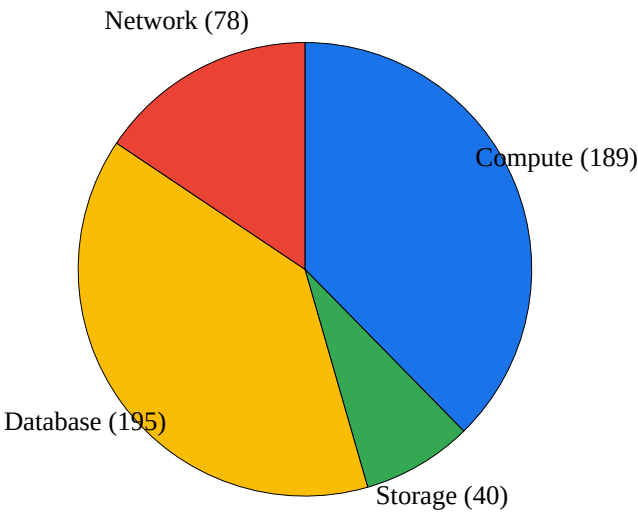
Recommended architecture for 'Retail Inventory API' in us-central1: 11 components spanning Backend, Backup, Database, Disaster Recovery, Load Balancer, Monitoring, Networking, Security, Storage.

Layer	Recommended Service	Rationale
Backend	Compute Engine (e2-standard-4 x 2)	Directly sized VM instances matching the requested/normalized compute profile.
Database	Cloud SQL (postgres, db-custom-2-8192) with High Availability (database failover)	Directly sized database instances matching the requested engine and sizing.
Load Balancer	External HTTP(S) Load Balancer	Distributes traffic across backend instances/nodes and terminates TLS.
Security	Cloud Armor	WAF and DDoS protection at the load balancer edge, recommended whenever a public LB is present.
Networking	Cloud CDN	Edge caching for static/cacheable content to reduce latency and origin load.
Storage	Persistent Disk (pd-balanced)	Block storage attached to compute instances.
Backup	Scheduled Persistent Disk Snapshots	Automated snapshots with 14-day retention.
Disaster Recovery	Multi-region backup replication	Cross-region snapshot/backup replication to support recovery objectives.
Networking	Multi-zone deployment	Spreads instances across zones within the region to survive a single-zone outage.
Monitoring	Cloud Monitoring + Cloud Logging	Standard observability stack for metrics, uptime checks, and centralized logs.
Security	IAM least-privilege roles + Secret Manager	Baseline access control and secrets management for any production workload.

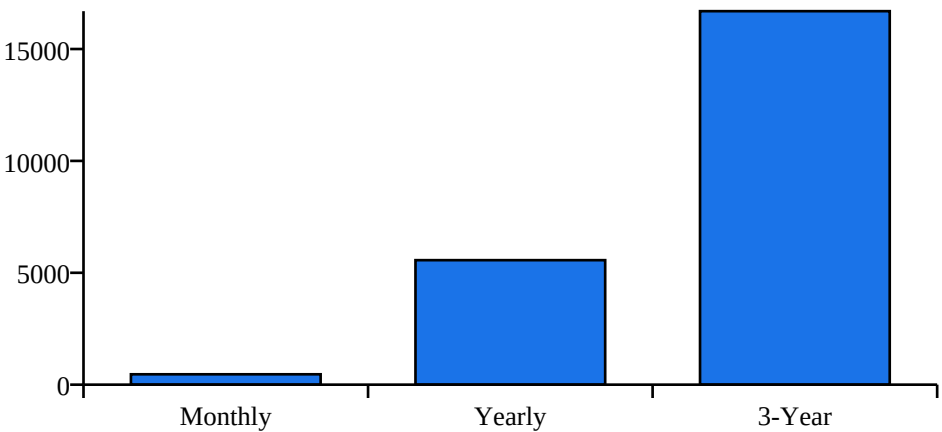
Itemized Pricing

Category	Description	Monthly Amount
Compute	e2-standard-4 x 2 instance(s) (us-central1)	USD 188.75
Storage	pd-balanced 157 GB x 2 disk(s) (us-central1)	USD 31.40
Storage	Persistent disk snapshots, 14-day retention	USD 8.16
Database	Cloud SQL db-custom-2-8192 (High Availability - synchronous standby) (us-central1)	USD 177.83
Database	Cloud SQL storage 100 GB (us-central1)	USD 17.00
Network	Internet egress, 500 GB/month	USD 60.00
Network	External HTTP(S) Load Balancer (us-central1)	USD 18.25

Cost Breakdown



Monthly vs. Yearly vs. 3-Year Cost



Assumptions

Field	Requested	Used	Reason
compute.vcpu	3	4	3 vCPU configuration does not exist in the e2 family. Selected nearest higher supported value (performance strategy).
compute.ram_gb	10.0	16.0	RAM is fixed per machine type in the e2 family; 'e2-standard-4' provides 16.0 GB.

Validation Results

Field	Severity	Reason
compute.vcpu	WARNING	3 vCPU does not exist as a standard configuration in the e2 family. Valid options: [1, 2, 4, 8, 16, 32].
compute.ram_gb	WARNING	10.0 GB RAM is not offered as a standalone configuration in the e2 family. Valid RAM values: [1.0, 2.0, 4.0, 8.0, 16.0, 32.0, 64.0, 128.0].
compute.machine_family	WARNING	No exact 'e2' machine type offers 3 vCPU with 10.0 GB RAM together.
region	INFO	'us-central1' is a supported Google Cloud region.
storage.size_gb	INFO	157 GB is within the valid range for 'pd-balanced'.
storage.snapshot_retention_days	INFO	Snapshot retention period is within a typical operational range.
database.vcpu_ram	INFO	Matches Cloud SQL tier 'db-custom-2-8192'.
network.load_balancer	INFO	Load balancer / CDN configuration is internally consistent.
network.estimated_egress_gb_per_month	INFO	Egress volume is within typical ranges.
availability.target_uptime_percent	INFO	Uptime target is achievable with the requested configuration.
availability.backup_frequency	INFO	Backup configuration is consistent with stated recovery requirements.

Savings Opportunities & Optimization Recommendations

- **Sustained Use Discount** is already applied, saving USD 37.75/month (20.0%).
- **compute.vcpu**: Normalize to the nearest supported vCPU count per the configured strategy.
- **compute.ram_gb**: Normalize to the RAM amount tied to the nearest supported vCPU/RAM pairing.
- **compute.machine_family**: Normalization engine will select the closest predefined machine type in this family, or an alternative family if none fits within tolerance.

Totals

Subtotal	USD 501.39
Discounts	-USD 37.75
Tax (0.0%)	USD 0.00
Support Plan (0.0%)	USD 0.00
Total Monthly	USD 463.64
Total Yearly	USD 5,563.68
Total 3-Year	USD 16,691.04

Approval

Prepared By:	FinOps Advisory Team	Date:	
Approved By:	Retail Client Inc.	Date:	